Ingeoexpert course
Numerical modelling of rock fracturing processes

Assignment 3

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Question 1

Fracture pattern at output step 40 (~1.12 ms or time step 80,000)

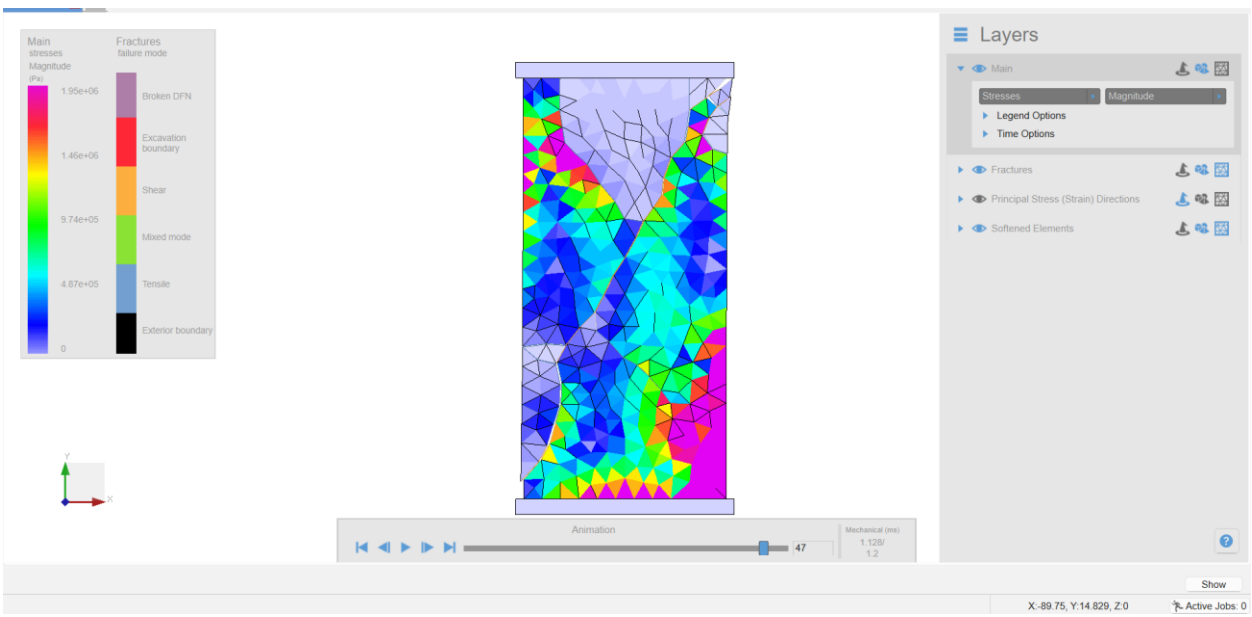


Fig. 1— Fracture pattern of the rock at ~1.128 ms, showing localized brittle failure within the rock sample under unconfined compression

The fracture pattern in Fig.1 shows that failure is no longer distributed uniformly throughout the sample but has localized into a dominant inclined damaged zone within the rock. The crack network is concentrated along a diagonal band extending across the sample, while several secondary cracks/fractures branch around the main failure path. This indicates that the specimen has progressed beyond the early crack-initiation stage and into a localized brittle failure regime. The concentration of damage away from a purely vertical splitting pattern also indicates that the specimen has moved beyond simple elastic behavior and has entered the stage of unstable crack growth and post peak failure.

Question 2

Stress magnitude plotted over time for an element around the centre of the specimen (hint: select the element, right click > Plot Over Time -> Plot Average of Selected Cells)

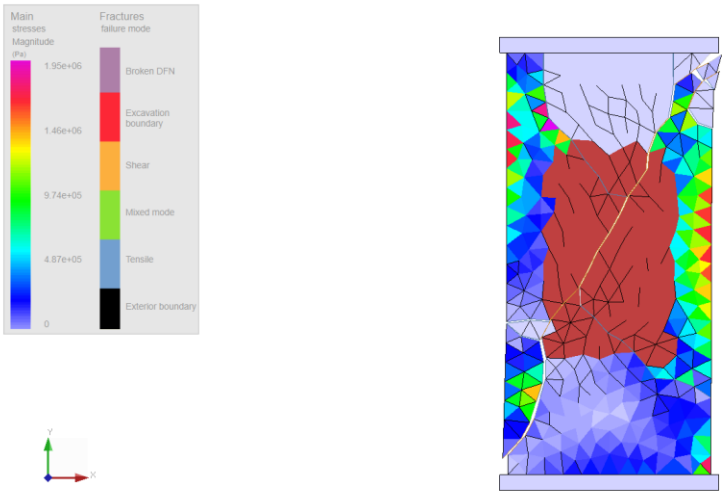


Fig. 2a— Visualization of stress magnitude over time for an element near the centre of the rock sample.

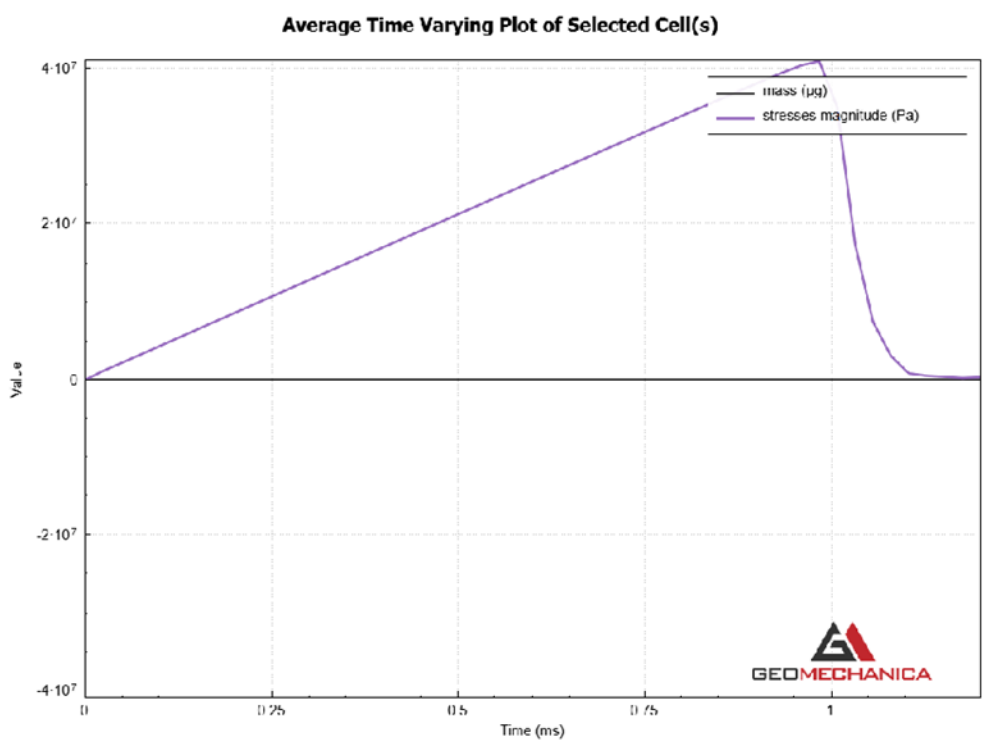


Fig. 2b— Chart of stress magnitude over time for an element near the centre of the rock sample.

The stress magnitude for the central element shows a clear increase with time during the loading phase, followed by a sharp reduction after reaching its maximum value. This trend indicates that the centre of the specimen initially carries increasing load as the platens compresses the rock, but once the main fracture zone develops and propagates through the sample, the selected element rapidly loses its load-bearing capacity.

Question 3

Force plotted over time for all nodes of the top platen

(hint: select ALL the nodes of the top platen, right click > Plot Over Time -> Plot Sum of Selected Points)

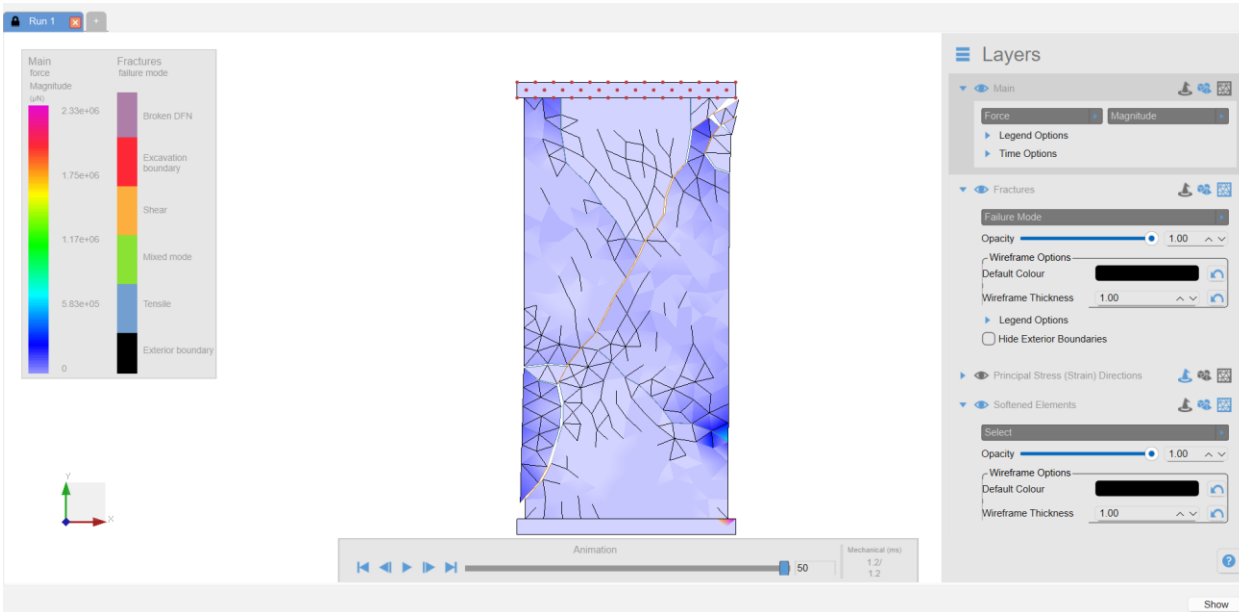


Fig.3a— Sum of nodal force on the top platen over time, showing increasing reaction force loading and a post peak drop after specimen failure

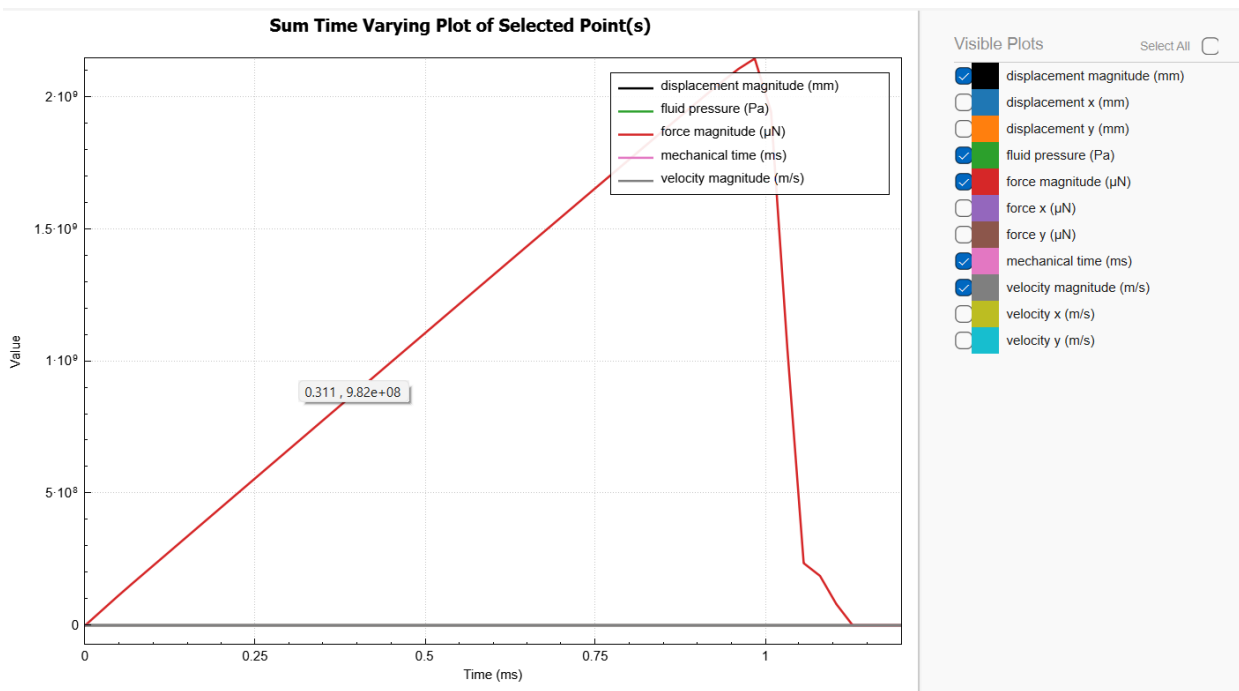


Fig.3b— Plot of nodal force magnitude on the top platen versus time, showing progressive load buildup during compression up to peak resistance, followed by a rapid post-peak drop after brittle specimen failure.

The summed force magnitude on the top platen increases steadily with time during the loading stage, indicating that the sample is resisting the applied platen displacement and accumulating compressive load. The reaction force reaches a maximum of about $1.0\,ms$, with a peak force magnitude of approximately $2.1\,kN$. After this maximum is reached, the curve drops sharply, indicating that the rock sample has its load-carrying capacity due to fracture localization and brittle failure.

Question 4

Using the built-in Python post-processing, show the sample's stress-strain curve and peak strength (UCS) in MPa (hint: see UCS chapter of the Tutorial Manual)

UCS = 42.77 MPa

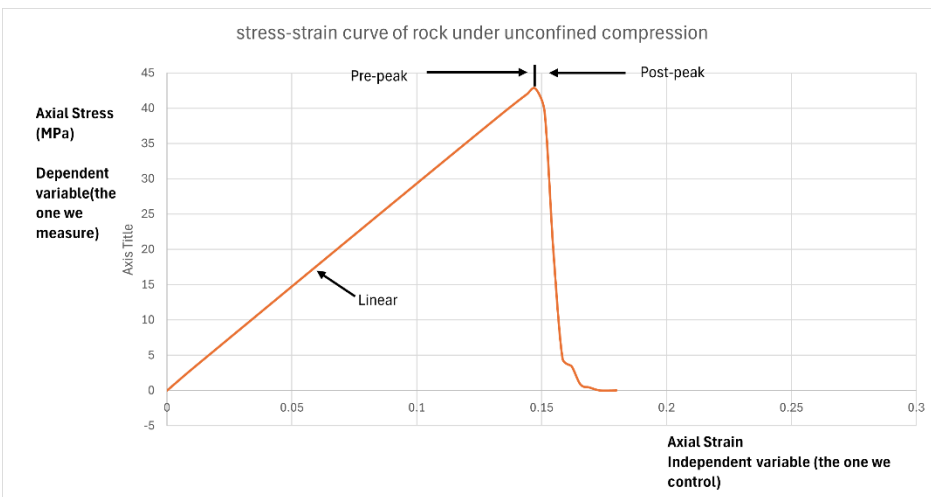


Fig. 4a— Complete stress-strain curve of rock under unconfined compression. The maximum stress that the rock could sustain (UCS) is 42.77 MPa (Processed in Excel Spreadsheet)

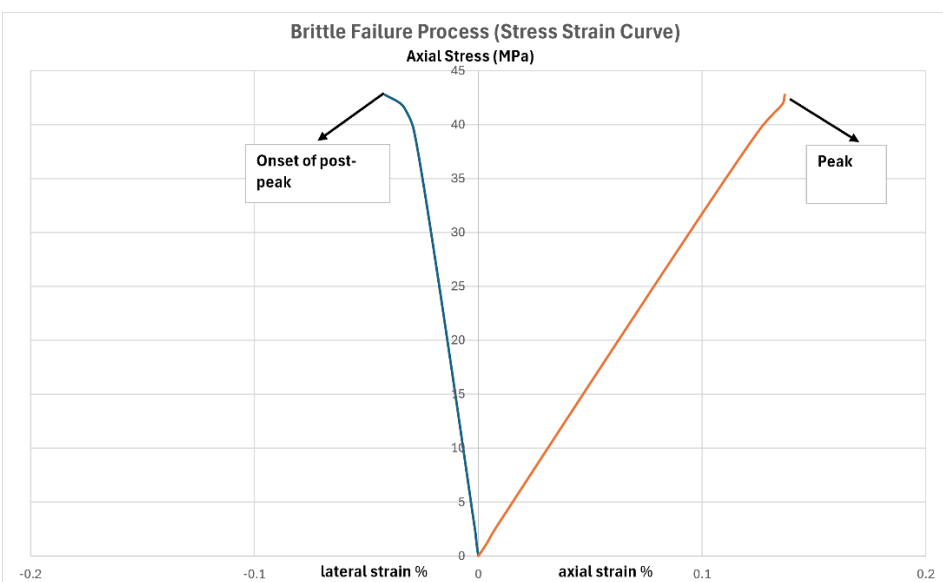


Fig. 4b— **Brittle failure process of rock (Processed in Excel Spreadsheet)**

Question 5

Description of failure pattern

The failure pattern in Fig.1 shows that failure is no longer distributed uniformly throughout the sample but has localized into a dominant inclined damaged zone within the rock. The crack network is concentrated along a diagonal band extending across the sample, while several secondary cracks/fractures branch around the main failure path. This indicates that the specimen has progressed beyond the early crack-initiation stage and into a localized brittle failure regime. The concentration of damage away from a purely vertical splitting pattern also indicates that the specimen has moved beyond simple elastic behavior and has entered the stage of unstable crack growth and post peak failure.

Question 6

Description of stress-strain curve

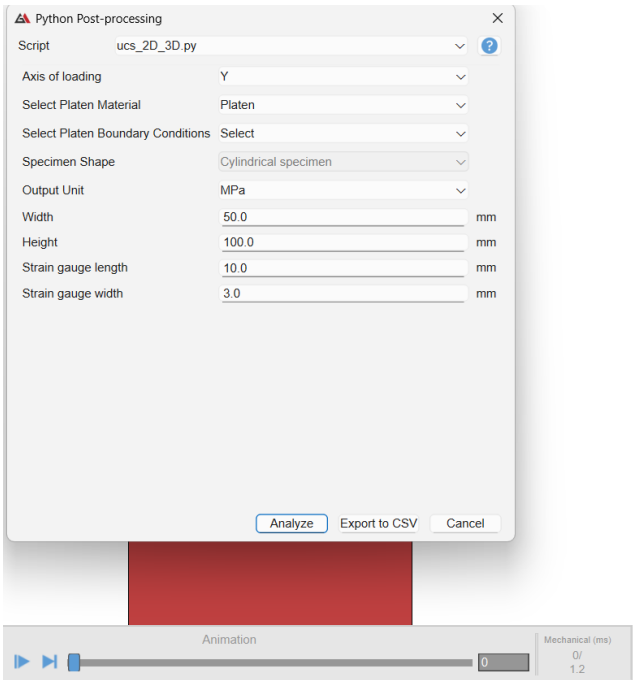
The stress strain curve as shown in Fig. 4a and b shows a characteristics behavior of rock samples subjected to uniaxial loading:

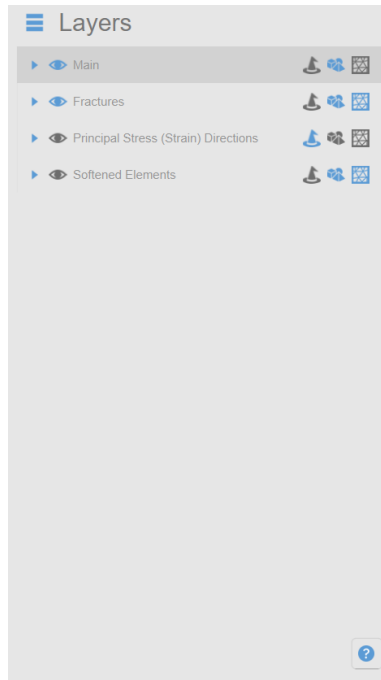
- Non-linear (concave upwards) response at low strains which is due to the closure of micro-cracks and initial lack of parallelism between loading platens.
- Linear response from which its slope can be used to determine the Young's modulus.
- Non-linear response (concave downwards) before peak which is due to intense microcracking.

The brittle failure process which is highlighted in fig.4b is characterized by 5 stages:

- Micro-crack closure and crushing of asperities
- Linear-elastic behaviour
- Onset of dilation and stable crack propagation
- Unstable crack growth
- Sample failure and brittle post-peak

Additional information:





Material Properties

Name	Platens	Rock
Density (kg/m3)	8000	2200
Damping Type	Viscous: Factor	Viscous: Factor
Viscous Damping Factor	1	1
Constitutive Law	Plane strain	Plane strain
Young's Modulus (Pa)	2.00E+11	3.00E+10
Poisson's Ratio	0.13	0.2
Friction Angle(°)	6	35
Fracture Model	Isotropic	Isotropic
Cohesion (Pa)	20E+6	10E+6
Tensile Strength (Pa)	5E+6	2.5E+6
Mode I fracture energy ($\mu N/mm$)	10	9400
Mode II fracture energy ($\mu N/mm$)	100	94000